

Low-Rank Adaptation for Scalable Large Language Models: A Comprehensive Survey

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May 01, 2025

Abstract

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Index Terms—large language models, multimodal learning, natural language processing, machine learning, representation learning, artificial intelligence

I. INTRODUCTION TO LOW-RANK ADAPTATION FOR LARGE LANGUAGE MODELS

The rapid advancement of large language models (LLMs) has brought new challenges in model adaptation [19], particularly regarding computational cost [29] and parameter efficiency [22]. Full parameter fine-tuning, illustrated in Figure 1, updates all model weights and demands substantial resources [27]. To address this, low-rank adaptation techniques most notably LoRA (Low-Rank Adaptation) [11] have been proposed to reduce the number of trainable parameters without

sacrificing performance. As shown in Figure 2, LoRA inserts low-rank matrices into weight updates, enabling efficient fine-tuning [7] while preserving the backbone model [40].

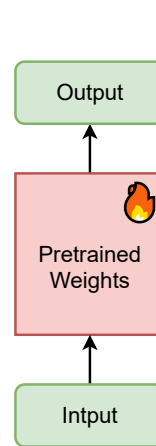


Fig. 1. Full-parameter fine-tuning.

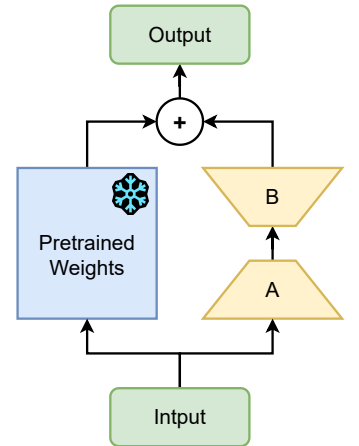


Fig. 2. LoRA: Low-Rank Adaptation of large language models.

Building upon LoRA, recent works have examined its theoretical foundations [5] [31] and practical extensions [38]. For instance, [9] provides an analytical perspective on LoRA’s behavior and limitations, without introducing new experiments, helping to clarify when and why low-rank adaptation works. Meanwhile, [17] introduces MoRA, a high-rank variant using square matrices rather than rank-constrained updates. As depicted in Figure 3, MoRA demonstrates stronger performance on memory-intensive tasks [28], highlighting that higher-rank

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updates can capture richer relationships in certain scenarios [48].

The extension of LoRA, as seen in [6], introduces dynamic adjustments to the intrinsic rank during adaptation through a gate unit, improving representation power while efficiently using parameters. Additionally, [50] updates not only low-rank matrices but also pre-trained weights using the delta of the product of two low-rank matrices [34], addressing limitations in learning representations for downstream tasks [35]. A comprehensive guide to low-rank approximation and adaptation is provided in [26], offering insights into their operation and effectiveness, along with new algorithms for future exploration. These papers share the common goal of achieving efficient adaptation of LLMs while maintaining or enhancing performance [11]. The widespread use of low-rank approximation and adaptation methods highlights their ability to reduce computational costs without significantly compromising model performance. However, contrasting viewpoints arise, such as the debate between low-rank and high-rank updating, with [17] advocating for high-rank updates to capture more complex relationships beneficial for certain tasks. Furthermore, [6] introduces dynamic rank adjustment, differing from the fixed rank approach in original LoRA and other methods, signaling a shift toward more flexible adaptation strategies.

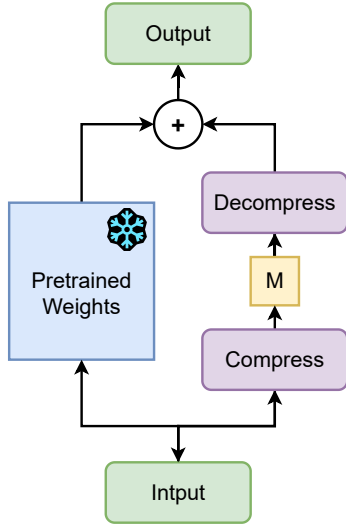


Fig. 3. MoRA: High-rank updating for parameter-efficient fine-tuning.

The technical details and methodologies employed in these papers demonstrate ongoing innovation in algorithm design for efficient fine-tuning [10]. For example, [50] explores specific mathematical formulations to update model weights, underscoring the importance of precise mathematical modeling in low-rank adaptation [44]. The introduction of gate units [42], square matrices [24], and delta updates highlights the creativity in designing algorithms to achieve parameter-efficient fine-tuning. These innovations illustrate the increasing focus on designing fine-tuning methods that optimize both performance and computational efficiency. In addition, [15] reveals that LoRA training in the neural tangent kernel (NTK)

[14] regime avoids spurious local minima, an important consideration in optimization. Another significant contribution comes from [33], which proposes MELoRA, as architecture shown in Figure 4, a mini-ensemble low-rank adapter that further enhances parameter-efficient fine-tuning by combining multiple low-rank adapters, improving the model’s robustness and generalization. These advancements offer new strategies for balancing fine-tuning efficiency with model performance in practical applications.

Nevertheless, several challenges remain, particularly in balancing efficiency and performance while ensuring the methods can adapt to various tasks. Fully understanding the mechanics and limitations of low-rank approximation in the context of large language models (LLMs) requires further research, as noted by [26]. Key issues include understanding the implications of low-rank adaptation on model interpretability and managing potential trade-offs between computational efficiency and model accuracy. Moreover, the connections between low-rank adaptation and other techniques, such as knowledge distillation, quantization, and other compression methods, remain areas of ongoing exploration, suggesting that the field is still evolving. These challenges indicate the complexity of optimizing low-rank methods in practice and underscore the need for continued innovation in this area to make the most of their potential in diverse applications.

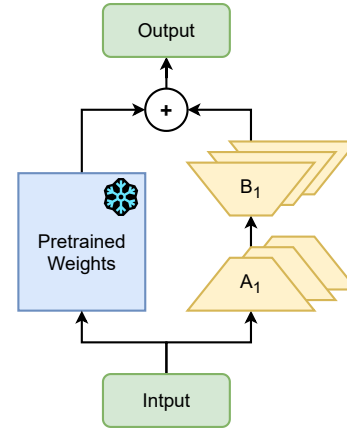


Fig. 4. MELoRA: Mini-ensemble Low-Rank Adapters for parameter-efficient fine-tuning.

The analysis of these studies underscores the diversity and rapid development of low-rank adaptation techniques for LLMs, with ongoing efforts to refine, extend, and understand these methods. As research continues, addressing the challenges and limitations identified will be crucial for advancing the field. The implications of this research are far-reaching, with potential applications in natural language processing, computer vision, and other areas where efficient model adaptation is critical. Ultimately, the development of more efficient and effective low-rank adaptation techniques will enable the widespread adoption of LLMs in a variety of applications, from speech recognition to text generation. Our contributions in this paper include:

- A comprehensive survey of recent low-rank adaptation techniques, including LoRA and its variants, highlighting their strengths, limitations, and potential for future improvements.
- An in-depth analysis of the trade-offs between parameter efficiency and model performance, providing insights into how these methods can be optimized for specific applications.
- A discussion on the emerging challenges and future directions in the field, emphasizing the need for further research in areas such as scalability, interpretability, and the integration of low-rank adaptation with other techniques.

II. BACKGROUND AND FOUNDATIONS OF LOW-RANK ADAPTATION TECHNIQUES

Low-rank adaptation (LoRA) has become a widely adopted method for efficiently fine-tuning large language models (LLMs), thanks to its simplicity and effectiveness [9]. Its applications now extend beyond language models to general-purpose foundation models, as reviewed in [45], where recent advances, core principles, and cross-domain use cases are systematically discussed. The underlying idea of low-rank approximation has a rich history in machine learning, signal processing, and natural language processing, as highlighted in [26], underscoring its foundational role in modern data analysis.

A central goal across these works is to reduce computational and memory overhead during fine-tuning, particularly for large models. To address the limitations of standard low-rank updates in LoRA, several new variants have been proposed. MoRA (High-Rank Updating for Parameter-Efficient Fine-Tuning) introduces a square matrix structure to achieve high-rank updates while keeping the number of trainable parameters constant [17]. In parallel, Delta-LoRA, which is shown in Figure 5, proposes a different approach: instead of updating only the low-rank matrices, it propagates learning to the pre-trained weights using the delta of the product of two low-rank matrices [50]. These innovations offer alternative perspectives on how to enhance adaptation flexibility without introducing substantial additional cost.

Across the literature, matrix decomposition and update strategies appear as recurring patterns, with LoRA, MoRA, and Delta-LoRA representing some of the most prominent methods. Their effectiveness has been validated in a variety of tasks, including instruction tuning, mathematical reasoning, and continual pretraining. Notably, these approaches consistently demonstrate improved efficiency while significantly reducing memory usage and computation. That said, the low-rank update mechanism is not without its drawbacks. MoRA’s introduction of high-rank updates is in part a response to LoRA’s performance bottlenecks in more complex settings [17]. Meanwhile, several studies have emphasized the need for clearer theoretical explanations and intuitive tools to help understand the behavior of low-rank approximations, particularly when applied to adaptation tasks [26]. Both practitioners and

researchers continue to explore methods that effectively trade off between efficiency and model expressiveness. Looking ahead, open challenges remain in scaling these techniques to even larger models, ensuring robustness across domains, and deepening our theoretical understanding of their mechanisms. Addressing these challenges will be essential for pushing the boundaries of what low-rank adaptation techniques can achieve in both research and practical applications [45].

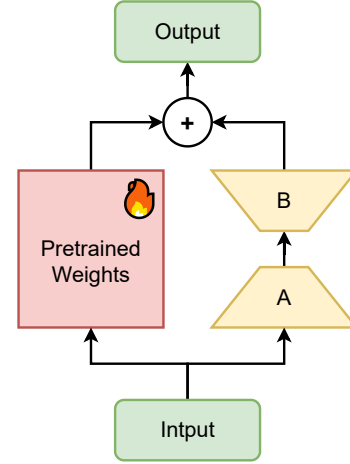


Fig. 5. Delta-LoRA: Fine-tuning high-rank parameters with the Delta of low-rank matrices.

The growing interest in low-rank adaptation techniques reflects their practical significance: they offer a way to fine-tune large language models more efficiently by reducing both computational burden and memory usage. These methods are not only relevant to model compression, but also closely tied to broader research themes such as transfer learning and meta-learning, where efficient model adaptation plays a pivotal role. Recent innovations—including MoRA and Delta-LoRA—further demonstrate the field’s momentum, each proposing alternative strategies to improve adaptability without sacrificing efficiency. Taken together, the foundational work in low-rank adaptation paints a comprehensive picture of the current research landscape and sets the stage for continued progress in building scalable, resource-aware fine-tuning solutions for large language models.

III. OVERVIEW OF LOW-RANK ADAPTATION METHODS FOR FINE-TUNING

Low-rank adaptation methods have become increasingly popular for fine-tuning large language models, owing to their ability to balance parameter efficiency and model effectiveness. One of the key insights from recent studies is that while LoRA, a widely adopted low-rank adaptation method, generally falls short of full fine-tuning in standard setups, it helps mitigate catastrophic forgetting and ensures that the base model retains performance on tasks outside the target domain [4]. On the other hand, MoRA, which employs high-rank updating, excels in memory-intensive tasks and achieves comparable results on other tasks by using a square matrix

to introduce higher-rank updates [17]. Meanwhile, ALoRA, which shown in Figure 8, offers a more flexible approach by dynamically adjusting the rank during the adaptation process, which leads to improved performance while keeping tunable parameters at similar levels [25]. LoRA-FA or LoRA-FB, which shown in Figure 6 and Figure 7, a memory-efficient fine-tuning method, optimizes memory usage by freezing the projection-down weight and updating the projection-up weight in each LoRA layer, all without sacrificing performance [47].

While low-rank adaptation methods are widely praised for reducing the number of trainable parameters, their overall effectiveness often hinges on more than just structural efficiency. In particular, memory usage remains a pressing concern, especially as model sizes grow. Techniques like LoRA-FA and MoRA have been developed to tackle this issue directly, offering ways to lower memory consumption without sacrificing performance. Alongside these efforts, researchers have also turned their attention to hyperparameter tuning, recognizing that the choice of parameters—such as rank size or learning rate—can significantly affect the final performance of low-rank methods [39]. In response, several studies have introduced optimization strategies tailored to the unique characteristics of low-rank adaptation. Beyond these core concerns, a number of recent approaches—such as GeLoRA’s use of geometric adaptive ranks [8], CoRA’s shared subspace optimization [43], and RoRA’s reliability-aware adaptation framework [23]—reflect a growing diversity in how researchers approach fine-tuning. Rather than relying on a one-size-fits-all solution, these developments indicate that low-rank adaptation is becoming more adaptable, with different strategies suited to different tasks.

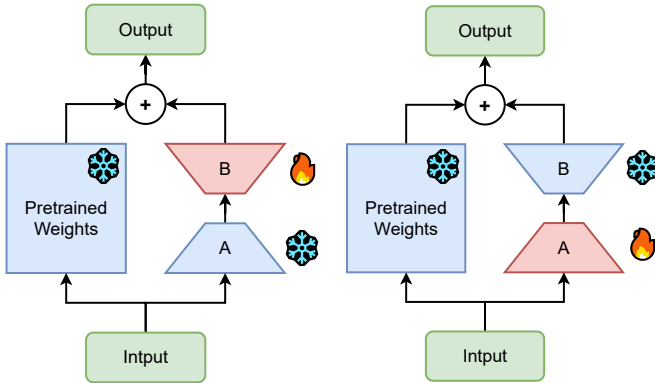


Fig. 6. LoRA-FA: Freeze model A. Fig. 7. LoRA-FB: Freeze model B.

The original LoRA paper [4] highlights the limitations of low-rank adaptation in standard settings, while MoRA [17] proposes a high-rank updating approach that challenges the traditional low-rank paradigm. ALoRA [25] takes a different approach by dynamically adjusting the intrinsic rank during the adaptation process, whereas LoRA-FA [47] focuses on reducing activation memory without modifying the underlying low-rank structure. The hyperparameter optimization paper [39] explores the use of blackbox optimization techniques,

which contrasts with the more traditional grid search or manual tuning approaches. Additionally, other papers have investigated the use of novel methods, such as Delta-tuning [4] and LoRA-based architectures [25], to improve the performance of low-rank adaptation methods.

Low-rank adaptation methods typically involve decomposing weight matrices into low-rank approximations using techniques like singular value decomposition (SVD) or randomized SVD. MoRA uses a square matrix to achieve high-rank updating, while ALoRA employs a novel method called AB-LoRA to estimate the importance score of each LoRA rank and dynamically prune abundant ranks [25]. LoRA-FA freezes the projection-down weight and updates the projection-up weight in each LoRA layer to reduce activation memory, whereas the hyperparameter optimization paper uses the NOMAD algorithm for blackbox optimization [39]. Other techniques, such as knowledge distillation and pruning, have also been explored to improve the efficiency of low-rank adaptation methods.

Low-rank adaptation methods may not always perform as well as full fine-tuning, particularly in standard settings [4]. The choice of hyperparameters, such as the rank of the decomposition, can significantly impact performance, making hyperparameter optimization a crucial step [39]. Memory efficiency remains a challenge for low-rank adaptation methods, with some approaches proposing solutions that may compromise performance or increase computational complexity. Furthermore, the dynamic adjustment of intrinsic rank during the adaptation process, as proposed by ALoRA, may require further research to fully understand its implications and potential applications.

Recent research in low-rank adaptation methods for fine-tuning large language models has yielded significant findings and advancements. The development of novel methods, such as MoRA, ALoRA, and LoRA-FA, has improved the efficiency and performance of low-rank adaptation methods. However, limitations and challenges remain, highlighting the need for further research to address these issues and advance the field of natural language processing. As the field continues to evolve, it is likely that low-rank adaptation methods will play an increasingly important role in enabling efficient and effective fine-tuning of large language models.

IV. TECHNICAL DEEP DIVE INTO LOW-RANK ADAPTATION ALGORITHMS

Recent advances in low-rank adaptation have significantly broadened the algorithmic toolkit available for efficiently fine-tuning large language models (LLMs). Building upon the foundational LoRA framework, which introduced the idea of injecting trainable low-rank matrices into frozen weight layers, subsequent research has sought to address its limitations in scalability, flexibility, and task adaptability. For instance, [9] reevaluates LoRA’s assumptions in large-scale deployment settings and emphasizes how practical constraints—such as memory, latency, and task-specific variations—demand more

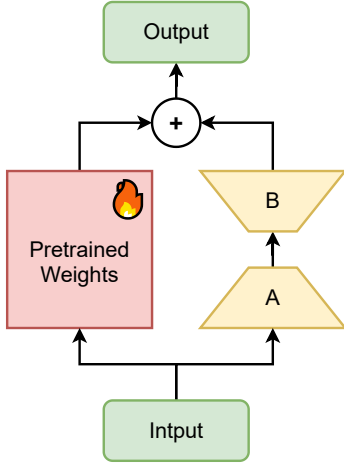


Fig. 8. ALoRA: Allocating Low-Rank Adaptation for fine-tuning large language models.

adaptable mechanisms. In parallel, a broader theoretical exploration of low-rank approximation methods is presented by [26], which positions low-rank structures not just as a memory-saving trick but as an inductive bias that shapes learning dynamics in high-dimensional models. Together, these perspectives suggest that low-rank adaptation is evolving from a fixed strategy into a design space encompassing multiple trade-offs between expressiveness, efficiency, and control.

This evolution is exemplified by the emergence of methods like SoRA [6] and Delta-LoRA [50], which introduce fundamentally different perspectives on how low-rank structures should interact with model updates. SoRA (Sparse Low-Rank Adaptation) incorporates a gate unit, trained via proximal gradient methods, that dynamically adjusts the rank of inserted matrices at each layer based on input features or gradient statistics. This flexibility enables targeted increases in model capacity for demanding tasks, while overall sparsity ensures efficient parameter use. In practice, SoRA introduces a sparsifying scheduler that encourages the retention of only those rank components deemed important, effectively combining the benefits of structured pruning with low-rank factorization. Delta-LoRA, in contrast, modifies not just the inserted low-rank matrices but also the original pre-trained weights. Its core update mechanism is defined as:

$$\Delta \mathbf{W}^{(t)} = \mathbf{A}^{(t+1)} \mathbf{B}^{(t+1)} - \mathbf{A}^{(t)} \mathbf{B}^{(t)}, \quad (1)$$

which is then added to the frozen base weights:

$$\mathbf{W}_{\text{new}} = \mathbf{W}_{\text{pretrained}} + \Delta \mathbf{W}^{(t)}. \quad (2)$$

This approach allows Delta-LoRA to propagate learning signals beyond the adapter alone, capturing temporal dynamics in the parameter space and avoiding the potential bottleneck of fixed low-rank modules. Notably, empirical results from [50] show that this formulation leads to faster convergence and better downstream task transferability, particularly in low-data regimes where fine-grained adjustments to base weights matter. These two methods reflect a growing interest in combining

low-rank decomposition with auxiliary mechanisms—gating, delta updates, scheduling—that give the model greater control over adaptation capacity.

One of the most pressing is the issue of scalability. While low-rank adaptation is computationally lighter than full fine-tuning, methods like SoRA introduce additional components—such as sparsity regularizers and gating networks—that increase the complexity of training. Their performance is often highly sensitive to hyperparameter choices and initialization, particularly when deployed on deep transformer stacks. Another area that remains underexplored is the theoretical grounding of these methods. Although intuitively appealing, it is still unclear how dynamic rank selection or delta-based updates influence the model’s generalization behavior or capacity control. As [26] notes, the lack of analytical tools to explain when and why low-rank adaptation works limits our ability to systematically improve these methods. Furthermore, ensuring robustness across varied datasets and domains—particularly under domain shifts or adversarial settings—remains a challenge. Addressing these questions will be crucial not only for improving existing algorithms but also for designing next-generation fine-tuning techniques that are both flexible and principled.

V. COMPARISON OF LOW-RANK ADAPTATION WITH FULL FINE-TUNING: PERFORMANCE AND TRADE-OFFS

The comparison between low-rank adaptation methods and full fine-tuning is crucial for assessing their practical applicability to large language models (LLMs). Low-rank methods like LoRA offer a significant reduction in the number of trainable parameters compared to full fine-tuning, making them attractive for applications where memory and computational resources are limited. However, as shown in [4], while LoRA performs well in certain contexts, it typically lags behind full fine-tuning in tasks requiring high expressiveness, where deeper and more task-specific adjustments to model weights are needed. On the other hand, LoRA excels at preserving the base model’s ability to generalize, particularly in scenarios where the model is exposed to out-of-domain data. Thus, LoRA serves as a conservative strategy that mitigates catastrophic forgetting while preserving general task performance.

The trade-off between model performance and parameter efficiency is particularly evident in the design of low-rank adaptation methods. As discussed in [23], increasing the rank in LoRA’s low-rank matrices can boost performance on target tasks, but this comes at the cost of increased computational overhead. This is particularly concerning in memory-constrained environments or applications where inference latency is critical. To address these issues, optimization strategies such as learning rate scheduling and scaling factor calibration, like those used in RoRA, have been developed to fine-tune LoRA’s scaling factor. These adjustments help mitigate the computational burden without sacrificing performance. Furthermore, [37] highlights a key observation: while LoRA’s weight matrices and those of fully fine-tuned models might exhibit similar task performance, their Singular Value

Decompositions (SVD) show distinct structural differences, which suggests that these two approaches converge to fundamentally different solutions.

Table I compares LoRA and full fine-tuning in terms of performance, memory usage, and adaptability, such as performance, memory usage, and adaptability to different data regimes.

In terms of practical deployment, context plays a decisive role in determining which method is more appropriate. For instance, LoRA has demonstrated superior performance in low-resource and cross-lingual scenarios. Its regularization effect prevents the model from overfitting to a small target dataset, enabling better generalization to new languages or tasks with limited data [41]. Conversely, when ample high-quality data is available and task-specific optimization is paramount, full fine-tuning can more effectively adapt the model to the target domain. Therefore, the choice between low-rank adaptation and full fine-tuning is not always straightforward but depends heavily on task complexity, data availability, and deployment constraints.

As the rank increases, LoRA’s computational and memory advantages begin to diminish, and performance gains start to plateau [23]. This trade-off, while manageable in many cases, underscores the need for further optimization. Methods like LoRA-FA [47], which optimize memory usage by freezing certain weights and recomputing activations, show promise but do not guarantee performance improvements in every case. Moreover, the spectral differences observed in LoRA’s weight matrices, as noted in [37], suggest that the solutions learned by LoRA and full fine-tuning may differ in fundamental ways, even when both models achieve comparable task performance. This points to a deeper need for theoretical insights into the underlying dynamics of low-rank adaptation and its relationship with model generalization.

While low-rank adaptation presents an attractive alternative to full fine-tuning, especially in scenarios requiring efficient resource usage and generalization across domains, its trade-offs must be carefully considered. The continued refinement of these methods, along with the development of new techniques to bridge the gap between efficient adaptation and task-specific performance, will be key to advancing the state of the art in large language model fine-tuning.

VI. ADVANCED LOW-RANK ADAPTATION TECHNIQUES: ROBUST ADAPTATION, SPARSE FINE-TUNING, AND HYBRID APPROACHES

Advanced low-rank adaptation techniques have emerged as a crucial area of research in the field of large language models, aiming to achieve efficient and effective fine-tuning while mitigating the challenges associated with full model updates. Recent studies, such as [30], have introduced novel methods like Robust Adaptation (RoSA), which combines low-rank and highly sparse components for parameter-efficient fine-tuning, outperforming existing approaches like LoRA and pure sparse fine-tuning in challenging tasks. This development highlights the trend towards exploring hybrid techniques that leverage the

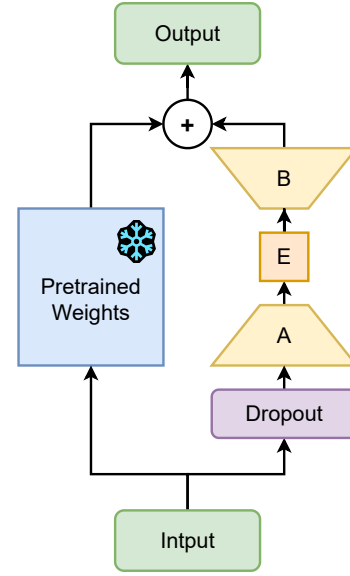


Fig. 9. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning.

strengths of both low-rank and sparse adaptations to enhance efficiency and performance.

The concept of dynamic rank adjustment has also been explored in works such as [6], which proposes Sparse Low-Rank Adaptation (SoRA), an extension of LoRA that adjusts the intrinsic rank during adaptation through a gate unit, offering more flexibility and potentially better performance with reduced parameters. Furthermore, techniques like CoRA, presented in [43], optimize LoRA training by leveraging shared knowledge in the form of a common subspace from large models, reducing unnecessary parameters and computational costs. These advancements underscore the significance of low-rank adaptation methods, such as LoRA, which have been shown to better maintain the base model’s performance on outside tasks and mitigate forgetting more than common regularization techniques [4].

A common theme among these papers is the pursuit of efficiency and effectiveness in adapting large language models. The exploration of sparse and hybrid approaches has become a focal point, with studies like [12] and [23] introducing efficient low-rank adaptation methods for sparse large language models and reliability optimization for rank adaptation, respectively. Additionally, works such as [36] have investigated robust and efficient fine-tuning of large language models with Bayesian reparameterization of low-rank adaptation, highlighting the importance of robustness and generalization in these techniques.

The trade-offs between full fine-tuning and low-rank adaptation have been a subject of discussion, with [4] contrasting the two approaches and highlighting the benefits of LoRA in maintaining base model capabilities. The choice between static and dynamic rank adjustment has also been explored, with SoRA’s dynamic approach offering a more flexible adap-

TABLE I
COMPARISON OF LORA VS. FULL FINE-TUNING.

Aspect	LoRA (and variants)	Full Fine-Tuning	Key Formula/Method
Trainable Parameters	$\sim 0.1\% - 5\%$	100%	$\mathbf{W}_{\text{LoRA}} = \mathbf{W}_{\text{pretrained}} + \Delta \mathbf{A} \mathbf{B}$
Performance (standard tasks)	Lower	Higher	Performance adjusted by rank size
Performance (low-resource / x-lingual)	Often better	Variable	LoRA regularization effect helps in small datasets
Memory Usage	Low	High	Efficiency achieved via low-rank matrices
Adaptation Flexibility	Modular	Full weight update	Modular updating of adapters
Update Target	Adapter only (or delta)	All weights	$\mathbf{W}_{\text{new}} = \mathbf{W}_{\text{pretrained}} + \Delta \mathbf{W}$
Generalization Outside Domain	Preserved	Often degraded	LoRA avoids overfitting with fewer parameters
Training Stability	High	Lower (risk of overfitting)	Robustness from reduced parameter updates

tation method [6]. Moreover, techniques like CoRA’s focused parameter reduction through shared knowledge differ from methods that focus on sparse or low-rank adaptations alone, demonstrating the diversity of approaches in this area [43].

Technical details and methodologies have also been extensively discussed, with RoSA utilizing robust principal component analysis for joint training of low-rank and highly sparse components [30]. SoRA employs a gate unit optimized with the proximal gradient method to dynamically adjust the rank, and introduces a sparsifying scheduler to examine parameter retention impact [6]. CoRA leverages a common subspace from large models to reduce LoRA’s parameters and computational needs [43]. These technical insights provide valuable guidance for the development and deployment of low-rank adaptation techniques.

One ongoing issue lies in the performance trade-offs: while these techniques are efficient in terms of parameter count and memory usage, they often lag behind full fine-tuning in certain high-complexity tasks. Choosing the right adaptation method remains non-trivial, as it depends on multiple factors including task specificity, dataset size, model architecture, and deployment constraints. Additionally, questions around scalability—particularly for very large models—and generalizability across domains and tasks continue to pose difficulties. For instance, although LoRA has been shown to mitigate catastrophic forgetting more effectively than traditional fine-tuning [4], the mechanisms behind this resilience are still not fully understood. Exploring how low-rank updates interact with internal model representations and how this can be controlled or enhanced remains an important direction for future research.

Looking forward, the development of more adaptive and theoretically grounded low-rank techniques is likely to shape the next phase of fine-tuning strategies for LLMs. The introduction of dynamically adjustable ranks, sparsity-aware schedules, and reliability-based adaptation—seen in recent methods such as SoRA and CoRA—signals a move toward more flexible and context-sensitive designs. As these methods continue to mature, they hold promise for scaling LLMs in a more sustainable manner, particularly in applications where rapid domain adaptation and resource efficiency are critical. Beyond model efficiency, the impact of low-rank adaptation extends into downstream tasks such as machine translation, summarization, and dialogue systems [6], [30], [43]. Ulti-

mately, the success of low-rank adaptation will depend not only on empirical gains, but also on our ability to integrate these techniques into robust, generalizable, and interpretable NLP systems. Continued exploration of this space is likely to yield both theoretical insights and practical innovations that push the boundaries of scalable language model adaptation.

VII. MODULAR AND PLUG-AND-PLAY LOW-RANK ADAPTATION FOR MIXED TASKS AND DYNAMIC SCENARIOS

Adapting large language models (LLMs) to diverse and evolving task demands requires methods that are not only parameter-efficient but also modular and flexible. Traditional full fine-tuning and even early low-rank approaches like LoRA often assume a static task context and fixed adapter structure, which limits their scalability in dynamic or multi-task settings. Recent research has focused on modular and plug-and-play low-rank adaptation methods, aiming to design components that can be selectively inserted, swapped, or composed based on task requirements. These methods seek to answer a central question: how can we adapt LLMs to varying input distributions and task complexities without retraining or duplicating the entire model? This shift from monolithic to compositional fine-tuning drives recent innovations in dynamic and meta-adaptive low-rank strategies.

Among the most prominent techniques, Dynamic LoRA [21] introduces a gating mechanism that adjusts adapter strength across transformer layers in response to both layer importance and input-specific characteristics. Instead of relying on static low-rank projections, it learns a gate $\gamma^\ell(\mathbf{x})$ that scales the adaptation term in each layer ℓ dynamically. This mechanism enables the model to activate or suppress low-rank adaptation conditionally, leading to more efficient allocation of adaptation capacity across layers and examples. In contrast, Delta-LoRA [50] reframes the update process itself by introducing a delta-based mechanism that captures the change in low-rank projections over consecutive steps. Rather than injecting a single adapter, it computes the temporal difference between low-rank factors $\mathbf{A}, \mathbf{B}$ at step t and $t+1$, and applies the resulting $\Delta \mathbf{W}$ directly to the base weights. This turns the adapter into a form of historical correction signal, effectively modeling how the adaptation evolves over training. As a result, Delta-LoRA enables structural modification of the pre-trained model in a parameter-efficient and incremental

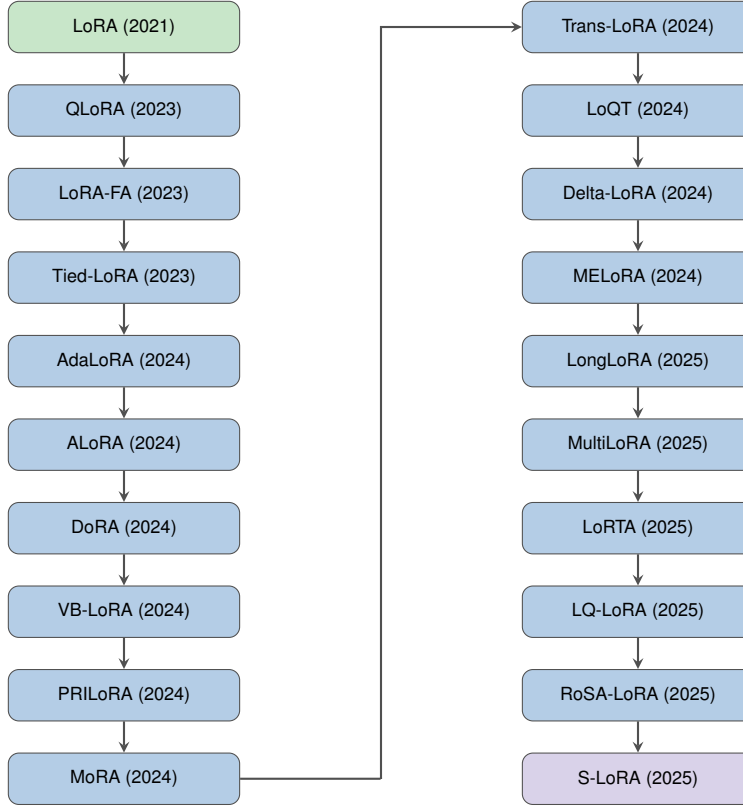


Fig. 10. Timeline of LoRA and its variants development. Each node represents a major innovation or extension in Low-Rank Adaptation methods for large language models.

manner, which contrasts with the purely modular insert-and-freeze approaches.

Other approaches explore the benefits of compositionality and parameter reuse by designing adapter mechanisms that can be selectively combined or shared across tasks. MELoRA [33], inspired by ensemble learning, introduces a set of lightweight low-rank adapters $\{(\mathbf{A}_i, \mathbf{B}_i)\}_{i=1}^K$, where each pair captures distinct subspaces of the model’s hidden representation. During inference, MELoRA computes a weighted combination of these adapters:

$$\mathbf{W}_{\text{MELoRA}} = \mathbf{W}_0 + \sum_{i=1}^K \alpha_i \mathbf{A}_i \mathbf{B}_i, \quad (3)$$

where the weights α_i are either learned or dynamically predicted based on the task or input. This formulation allows MELoRA to maintain a high effective rank through composition while keeping the total number of parameters low. It also introduces the flexibility to emphasize or suppress specific components depending on task requirements.

CoRA [43], on the other hand, adopts a shared subspace perspective by decomposing the task-specific adaptation into a common basis projection. It assumes that the task adapters lie in a global latent space parameterized by matrices $\mathbf{U}, \mathbf{V}$ shared among all tasks. For a specific task t , CoRA learns a task-specific scaling matrix $\mathbf{S}_t$ and defines the adaptation as:

$$\mathbf{W}_t = \mathbf{W}_0 + \mathbf{U} \mathbf{S}_t \mathbf{V}^\top. \quad (4)$$

This structure encourages alignment between tasks by enforcing low-rank transformations within a constrained subspace, improving generalization in multitask or few-shot settings. Compared to MELoRA’s adapter ensemble, CoRA promotes cross-task efficiency and compatibility through shared representational geometry.

To compare the design philosophy and capabilities of these methods, Table II summarizes key characteristics in several dimensions, including parameterization, update strategy, task sensitivity, and modularity. The inclusion of representative equations provides further insight into the underlying adaptation mechanisms.

Although methods are becoming increasingly sophisticated, several core challenges persist. To begin with, current approaches still depend heavily on empirical heuristics for adapter placement and selection. Even though Dynamic LoRA incorporates input-dependent gating, the criteria for adapter activation or sharing remain largely opaque. In addition, balancing generalization with task specificity proves difficult—methods like CoRA attempt to learn a shared latent space, but such abstractions may dilute essential task-specific signals. Furthermore, the computational advantages of plug-and-play methods can erode as the number of adapters grows

TABLE II
COMPARISON OF MODULAR LOW-RANK ADAPTATION METHODS.

Method	Update Mechanism	Task Sensitivity	Modularity	Key Formula
LoRA [9]	Static low-rank insert	No	Fixed module	$\mathbf{W} = \mathbf{W}_0 + \mathbf{A}\mathbf{B}$
Dynamic LoRA [21]	Layer/input-dependent gate	Yes	Conditional insert	$\mathbf{W} = \mathbf{W}_0 + \gamma(\mathbf{x})\mathbf{A}\mathbf{B}$
Delta-LoRA [50]	Updates base weights via delta	Moderate	Semi-fixed	$\Delta\mathbf{W} = \mathbf{A}^{t+1}\mathbf{B}^{t+1} - \mathbf{A}^t\mathbf{B}^t$
MELoRA [33]	Mini-ensemble adapters	Yes	Highly modular	$\mathbf{W} = \mathbf{W}_0 + \sum_i \alpha_i \mathbf{A}_i \mathbf{B}_i$
CoRA [43]	Shared subspace projection	Indirect	Shared plug-in	$\mathbf{W}_{\text{task}} = \mathbf{U}\mathbf{S}_t\mathbf{V}^\top$

or their interactions become unpredictable. Optimization in these compositional settings is also insufficiently explored, particularly under constraints such as continual learning or memory-limited inference. Lastly, the theoretical foundations for why and when modular low-rank adaptation is effective remain underdeveloped, with limited clarity on its relationship to model capacity, transferability, or representation alignment.

A promising future direction lies in the unification of modular components under a controller framework capable of learning when to activate, combine, or bypass specific adapters based on input features and task demands. Such a controller could be trained to balance performance with efficiency, ideally using self-supervised signals or proxy objectives. Additionally, there is growing interest in extending modular low-rank adaptation to multimodal or multilingual settings, where different adapters could correspond to language or modality-specific priors. These directions suggest that the future of low-rank adaptation may not lie in designing ever more sophisticated adapters, but in orchestrating simpler ones more intelligently.

VIII. EFFICIENT IMPLEMENTATION AND SYSTEM SUPPORT FOR LOW-RANK ADAPTATION

The practical deployment of low-rank adaptation (LoRA) techniques for large language models (LLMs) increasingly demands solutions that extend beyond algorithmic innovation into system-level efficiency and adaptability. As LLMs continue to scale in size and application domains, the tension between fine-tuning flexibility and deployment constraints becomes more pronounced. Static low-rank designs, such as standard LoRA, provide a strong baseline for reducing trainable parameters, but often fall short when facing heterogeneous downstream tasks, limited hardware budgets, or dynamic user inputs. To bridge this gap, recent research has focused on efficient implementations that allow for task-aware, sparsity-controlled, and rank-adaptive adaptation within a unified system interface.

A central theme in this area is the move from fixed-rank insertion to dynamic and sparsity-aware rank scheduling. In particular, Sparse LoRA (SoRA) [6] introduces a learned gate γ_i for each adapter unit, which modulates the contribution of the low-rank projection:

$$\mathbf{W}_{\text{SoRA}} = \mathbf{W}_0 + \sum_i \gamma_i \mathbf{A}_i \mathbf{B}_i, \quad (5)$$

where $\gamma_i \in [0, 1]$ is optimized to induce rank sparsity. This formulation enables not only parameter reduction but also structural sparsity, which can be exploited by inference-time optimizers or compiler backends for further acceleration. The gate values are trained via proximal gradient descent, transforming the adapter learning process into a sparse optimization problem. Similar goals are shared by ALoRA [25], which adjusts the intrinsic rank dynamically based on layerwise sensitivity, leading to flexible rank allocation that adapts to the needs of different components in the model.

Building on this concept of differentiated adaptation, RankAdaptor [49] integrates system-level performance modeling into the rank assignment process. Instead of relying on heuristics or static rules, it solves an optimization problem of the form:

$$\mathbf{r}^* = \arg \min_{\mathbf{r}} \mathcal{L}_{\text{task}}(\mathbf{r}) + \lambda \cdot \mathcal{C}_{\text{model}}(\mathbf{r}), \quad (6)$$

where $\mathbf{r}$ denotes the rank configuration across layers, $\mathcal{L}_{\text{task}}$ captures task-specific loss, and $\mathcal{C}_{\text{model}}$ represents a system performance model estimating memory or latency. This framework allows hierarchical rank allocation to emerge as a byproduct of joint optimization, with fine-tuning efficiency dictated not just by task performance but also by system-level constraints. The method has proven especially useful for adapting pruned LLMs, where different layers exhibit varying sensitivity to additional parameters.

System-level implementations of these ideas often rely on efficient runtime support to handle sparse and modular adapter structures. For instance, dynamic LoRA variants require input-aware gating computation at each forward pass, which poses challenges for batching and kernel fusion. Recent efforts such as performance-aware scheduling [49] and hardware-friendly adapter design [3] explore compiler-level optimizations and runtime-aware training strategies to maintain throughput. Moreover, integration with parameter-efficient fine-tuning libraries like HuggingFace’s PEFT or DeepSpeed’s ZeRO-Inference shows that real-world deployment increasingly depends on well-designed adapter interfaces, low-overhead parameter storage, and compatibility with mixed-precision inference.

Several challenges remain open. The balance between adaptation granularity and runtime overhead is non-trivial—finer control over rank or sparsity may introduce more dynamic computation that hinders GPU utilization. Furthermore, while methods like SoRA reduce parameter counts, the actual mem-

ory footprint can vary depending on implementation, especially in sparse or quantized formats. From a deployment perspective, one pressing issue is adapter composability: how to switch, stack, or blend multiple adapters without reloading full models or disrupting memory layouts. These challenges highlight the need for system-level frameworks that jointly consider algorithm design, runtime support, and hardware constraints.

Future research is likely to move toward unifying adaptation strategies with system schedulers and compilers. One potential direction is the development of controller modules that dynamically determine adapter layout and activation based on runtime feedback, user queries, or task identity. Another is to design unified adapter graphs that support multi-task, multilingual, or multi-domain adaptation through routing policies over modular low-rank blocks. As LLMs are increasingly deployed across diverse settings—from edge devices to cloud-based APIs—the success of LoRA will depend not only on its parameter efficiency but also on its ability to align with the infrastructure on which it runs. In this regard, efficient LoRA is no longer just an algorithmic goal, but a system-level contract between adaptability and deployability.

IX. APPLICATIONS OF LOW-RANK ADAPTATION IN REAL-WORLD SCENARIOS: CHALLENGES AND OPPORTUNITIES

Low-rank adaptation techniques have increasingly been applied to large-scale language models (LLMs), especially in real-world tasks where computational efficiency and model performance must be carefully balanced. One of the most successful methods is LoRA [9], which has become a popular choice for fine-tuning pre-trained LLMs due to its simplicity and effectiveness. LoRA significantly reduces the number of trainable parameters while retaining performance across a range of tasks. However, challenges arise when applying LoRA to larger models, where fixed ranks and simple adapter designs may not be sufficient for task-specific adaptation. In this context, methods like MoRA [17] introduce high-rank updates while maintaining the same number of trainable parameters, offering a more flexible approach to model adaptation. This raises an ongoing debate between low-rank and high-rank methods, with MoRA showing potential for more complex tasks that require a richer adaptation space.

A shift toward dynamic low-rank adaptation has led to the development of methods such as SoRA [6], which allows for rank adjustments during training based on the complexity of the task and input data. This dynamic approach enables sparse updates and significantly reduces memory usage while maintaining high performance. Similarly, MELoRA [33] introduces a mini-ensemble of adapters, enabling the model to adapt to various tasks by selecting a weighted combination of low-rank adapters. This method enhances cross-task generalization, offering a more modular approach to fine-tuning without overburdening the model with a large number of trainable parameters.

As LLMs grow in size and complexity, methods like LoRA and MoRA struggle to maintain performance across larger models and datasets. Moreover, low-rank methods still face limitations in terms of adaptability to a wide variety of tasks, particularly when it comes to real-time, dynamic tasks where input characteristics can change rapidly. To address these issues, future research should focus on better integrating low-rank adaptation with hardware-aware scheduling, as well as developing self-adaptive rank allocation methods that can scale with both model size and task complexity.

The application of low-rank adaptation techniques extends far beyond NLP tasks. Their use in computer vision, multimodal learning, and other areas where large models are required for fine-grained adaptation continues to grow. The challenges of scaling, task-specific customization, and dynamic adaptability are not only technical but also system-level, requiring advances in training frameworks, hardware optimizations, and inference pipelines. By addressing these challenges, low-rank adaptation will play a pivotal role in the future of efficient and effective machine learning, enabling the widespread deployment of LLMs across diverse applications.

X. ANALYSIS OF LEARNED SOLUTIONS: SPECTRAL PROPERTIES, GENERALIZATION BEHAVIORS, AND INTRUDER DIMENSIONS

Low-rank adaptation methods, such as LoRA, have become key to fine-tuning large language models, offering efficient solutions by reducing the number of trainable parameters. However, spectral analysis reveals that LoRA and full fine-tuning yield fundamentally different weight matrices, which has profound implications for their generalization behavior [37]. LoRA-trained models, though achieving comparable performance in some tasks, may not generalize as well to tasks outside the distribution of the training data due to the introduction of "intruder dimensions." These dimensions, which do not correspond to task-relevant features, can hinder performance on out-of-domain tasks. This highlights a key trade-off: while LoRA reduces the number of trainable parameters, it may sacrifice some generalization ability, especially when compared to full fine-tuning methods that utilize the full spectrum of model weights.

The analysis of generalization error using spectral properties provides a framework for understanding how model behavior changes with different adaptation strategies. Notably, LoRA's approach, operating in the neural tangent kernel (NTK) regime, avoids spurious local minima, facilitating gradient descent's ability to converge to low-rank solutions [15]. This theoretical support reinforces the practical effectiveness of LoRA, which is also seen in related techniques like LaMDA and MELoRA, where significant parameter reductions are achieved while maintaining or even improving performance [2], [33].

The impact of these dimensions is still a topic of active research, with potential solutions including dynamic rank adjustments and ensemble methods like MELoRA, which introduce additional complexity but offer enhanced generalization

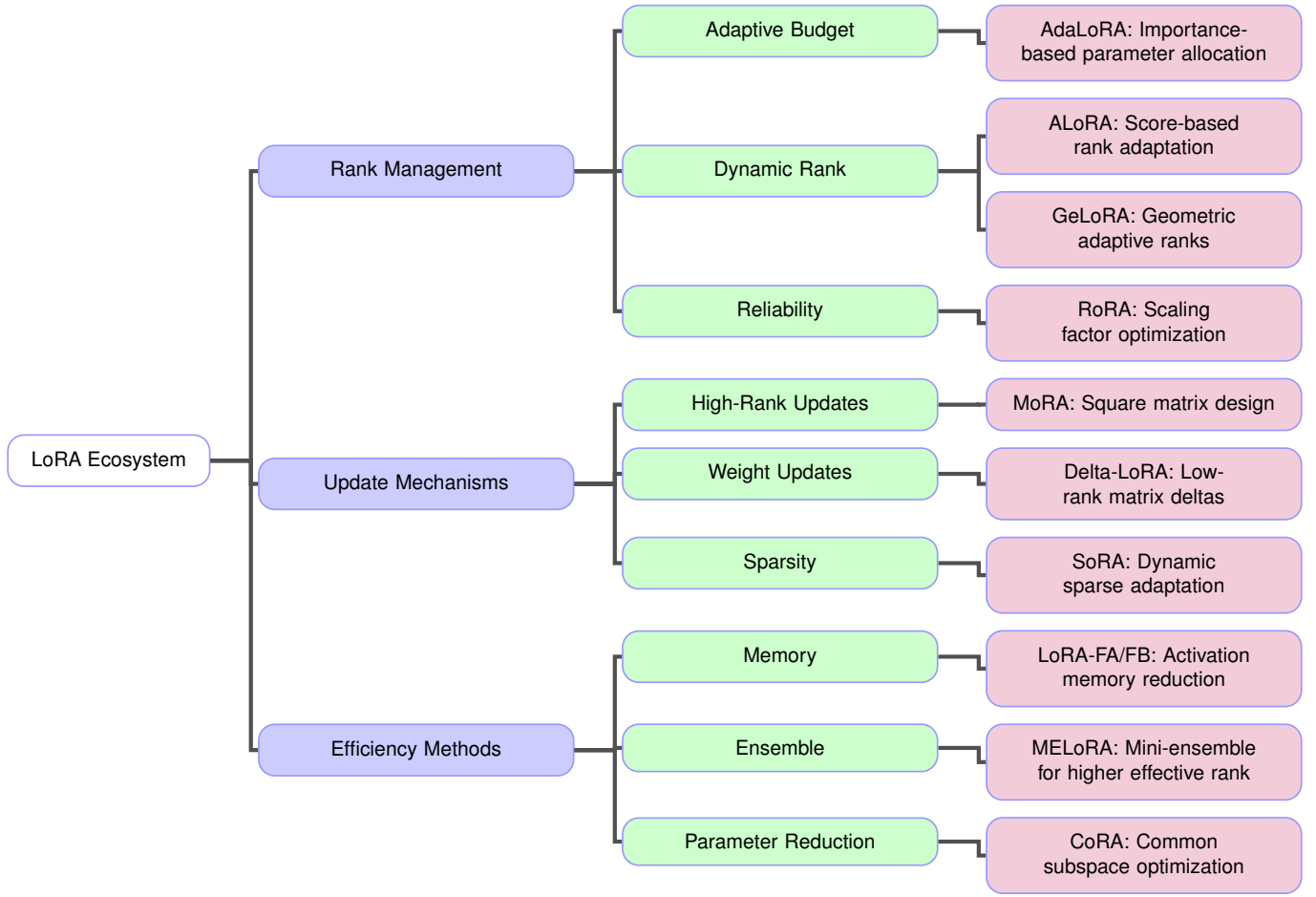


Fig. 11. Taxonomy of Low-Rank Adaptation methods for large language models.

capabilities. However, these approaches come with increased computational costs [20], [33].

Moreover, hyperparameter tuning, especially of the rank and learning rate, is a critical challenge in fine-tuning LoRA models. Finding the optimal balance between rank size, training time, and generalization performance remains an open problem. As model size increases, so does the complexity of this balance, necessitating the development of more robust algorithms that can automatically adjust these parameters during training [2], [33]. Future research will likely focus on optimizing these hyperparameters to minimize the effects of intruder dimensions and enhance the scalability and generalization of low-rank adaptation techniques.

Low-rank adaptation methods like LoRA represent a promising approach to efficiently fine-tuning large language models, but challenges remain in their application to complex, real-world tasks. The trade-offs between efficiency, generalization, and computational cost highlight the importance of continued research into spectral properties, task-specific adaptations, and the effects of intruder dimensions. By addressing these issues, the full potential of low-rank adaptation can be realized, enabling more efficient and scalable fine-tuning for a wide range of applications [2], [15], [37].

XI. FUTURE DIRECTIONS AND OPEN CHALLENGES IN LOW-RANK ADAPTATION RESEARCH

The field of low-rank adaptation research has witnessed significant advancements in recent years, with a plethora of techniques being proposed to efficiently fine-tune large language models [9], [45]. A comprehensive review of low-rank adaptation techniques for foundation models is provided in [45], which discusses recent techniques, emerging frontiers, and applications across multiple domains. The Ensembles of Low-Rank Expert Adapters (ELREA) framework, proposed in [20], improves model capability in handling diverse tasks by clustering training instructions based on gradient directions and utilizing low-rank adaptation for efficiency. Furthermore, [26] seeks to clarify the mechanics of low-rank approximation and its application in adaptation, providing a comprehensive guide that reveals their inner workings and explains their utility.

A common theme emerging from the analyzed papers is the emphasis on efficiency and scalability, with low-rank adaptation techniques being recognized for their ability to provide parameter-efficient fine-tuning [17], [6]. The need for adaptability and generalization is also a recurring motif, with papers focusing on improving model adaptability to specific

downstream tasks and enhancing generalization across tasks [41], [15]. Low-rank approximation and its applications are recognized as fundamental in modern data analysis, with applications extending beyond natural language processing to other domains like computer vision and scientific discovery [8], [25].

However, contrasting viewpoints also exist, such as the debate between low-rank vs. high-rank updating [17], [50]. While LoRA utilizes low-rank updating, MoRA introduces an alternative approach with high-rank updating for potentially better knowledge acquisition and memorization. Additionally, ensemble methods such as ELREA contrast with traditional single-adapter methods, suggesting that handling task diversity might require more complex strategies [33], [13]. The technical details of note include the mathematical underpinnings of the low-rank approximation and adaptation, aiming to provide a deeper understanding [1], [18], as well as algorithmic innovations such as ELREA and MoRA, which offer alternative or complementary approaches to existing methods such as LoRA.

Limitations and challenges persist, including scalability and robustness [46], theoretical understanding [49], task diversity and generalization [32], and balancing efficiency and effectiveness [16]. The need for deeper theoretical insights into how low-rank approximation and adaptation operate, especially in complex scenarios, is acknowledged [9]. Effectively handling diverse tasks without compromising model generalization is a persistent challenge, and striking a balance between the parameter efficiency of low-rank adaptation methods and their ability to learn and memorize new knowledge effectively is an ongoing research direction [45], [20].

Low-rank adaptation research has made significant strides in recent years, with a focus on efficient fine-tuning of large language models. Key themes and developments include the emphasis on efficiency and scalability, adaptability and generalization, and the recognition of low-rank approximation as a fundamental technique in modern data analysis. However, limitations and challenges remain, highlighting the need for continued research into this field to address the ongoing challenges and unlock the full potential of low-rank adaptation techniques [26], [8]. Ultimately, the implications of low-rank adaptation research are far-reaching, with connections to various domains and applications, underscoring the importance of continued innovation and exploration in this area [45], [20].

XII. OUTLOOK FOR LOW-RANK ADAPTATION IN LARGE LANGUAGE MODELS

The field of low-rank adaptation (LoRA) has demonstrated remarkable progress in improving the efficiency and scalability of large language models (LLMs), providing a promising direction for adapting pre-trained models to diverse tasks. LoRA’s simplicity and effectiveness in reducing trainable parameters while maintaining performance have made it a widely adopted method in NLP and beyond [9]. Recent studies have solidified the theoretical underpinnings of LoRA, particularly in the context of the neural tangent kernel (NTK) regime,

where LoRA eliminates spurious local minima, enhancing its ability to generalize [15].

The success of LoRA has led to the development of several extensions aimed at improving its representational power and adaptability. MoRA [17] introduces high-rank updates, challenging the low-rank framework by offering a richer adaptation space. SoRA [6], on the other hand, introduces dynamic rank adaptation, allowing for more flexible updates. Delta-LoRA [50] enhances LoRA by fine-tuning high-rank parameters with the delta of low-rank matrices, which has shown improvements on tasks requiring higher memorization or handling large amounts of data. These innovations demonstrate that while LoRA’s low-rank approach is effective, high-rank and dynamic adaptation methods can improve performance, particularly in tasks that require greater capacity for memorization.

However, a key trade-off remains between parameter efficiency and model expressiveness. While LoRA and its variants reduce the number of trainable parameters, they sometimes compromise generalization ability, especially when scaling to more complex tasks. Higher-rank models like MoRA can achieve better performance in such cases, but at the cost of increased computational requirements. Methods such as SoRA address this by dynamically adjusting the rank, offering a compromise between efficiency and expressiveness without significantly increasing computational overhead.

The challenge of scaling these methods to larger models or more complex tasks is a recurring theme. While LoRA can scale effectively to medium-sized models, methods like MELoRA [33], which use ensembles of low-rank adapters, provide a more flexible approach for cross-task generalization, though this incurs additional computational costs. Future research will need to address these scalability challenges, particularly in multi-task learning or applications requiring real-time adaptation. Additionally, advancements in spectral learning, adaptive rank allocation, and hardware-aware optimization will be crucial for further enhancing the scalability and efficiency of low-rank adaptation techniques.

The future of low-rank adaptation in large language models is promising, with ongoing developments offering more powerful and flexible methods for efficient fine-tuning. As the field evolves, new methods that integrate adaptive mechanisms, such as dynamic rank adjustment and ensemble models, will likely continue to improve performance across diverse tasks. The combination of efficiency, scalability, and adaptability positions low-rank adaptation as a key technique in the ongoing advancement of machine learning, enabling the deployment of more capable and resource-efficient models across a wide range of applications.

XIII. CONCLUSION

In conclusion, low-rank adaptation (LoRA) has emerged as an efficient method for fine-tuning large language models (LLMs), reducing parameter complexity while maintaining performance. However, the limitations of LoRA, such as its inability to fully capture task-specific knowledge, have led to the development of alternative approaches like MoRA

TABLE III
COMPARISON OF CORE PERFORMANCE CHARACTERISTICS.

Method	Performance on Specific Tasks	Adaptability	Computational Efficiency	Generalization Ability
LoRA [9]	High (for specific tasks)	Moderate (fixed rank)	Low (static rank, limited updates)	Moderate (may struggle with complex tasks)
MoRA [17]	Very High (with large ranks)	High (for large tasks)	Moderate (due to high rank)	High (multi-task performance)
SoRA [6]	High (dynamic rank)	Very High (adaptive rank)	Very High (efficient sparsity control)	High (dynamic adaptation)
Delta-LoRA [50]	Moderate (lower rank)	Low (base weight update limitation)	Moderate (requires adaptive learning rate)	Moderate (on out-of-domain tasks)
MELoRA [33]	Very High (ensemble)	High (task-specific adaptability)	High (ensemble computation cost)	Very High (cross-task generalization)

TABLE IV
COMPARISON OF METHOD CHARACTERISTICS AND CHALLENGES.

Method	Spectral Properties	Intruder Dimensions Impact	Computational Cost	Scalability Challenges
LoRA [9]	Low-rank decomposition	Moderate (task-dependent)	Low (fixed rank)	Low (easy to scale to medium models)
MoRA [17]	High-rank decomposition	Low (due to non-adaptive rank)	Moderate (requires larger ranks)	Moderate (larger models require more rank)
SoRA [6]	Dynamic rank (sparsity)	Low (adaptive rank control)	Low (optimized gradients)	High (scalability with task complexity)
Delta-LoRA [50]	Incremental low-rank updates	High (due to weight updates)	Moderate (requires adaptive learning rate)	Moderate (needs additional memory for delta updates)
MELoRA [33]	Ensemble of low-rank adapters	High (ensemble complexity)	High (ensemble parameter cost)	High (ensemble complexity)

and Delta-LoRA, which introduce higher-rank updates and dynamic adjustments. Despite these advancements, challenges such as scalability, task-specific performance variations, and the need for a deeper theoretical understanding remain. Future research should focus on enhancing the scalability, efficiency, and adaptability of low-rank methods while addressing these challenges to optimize LLM fine-tuning for diverse applications.

REFERENCES

- [1] Afia Anjum, Maksim E Eren, Ismael Boureima, Boian Alexandrov, and Manish Bhattarai. Tensor train low-rank approximation (tt-lora): Democratizing ai with accelerated llms. *arXiv preprint arXiv:2408.01008*, 2024.
- [2] Seyedarmin Azizi, Souvik Kundu, and Massoud Pedram. Lamda: Large model fine-tuning via spectrally decomposed low-dimensional adaptation. 2024.
- [3] Daniel Bershtatsky, Daria Cherniuk, Talgat Daulbaev, Aleksandr Mikhalev, and Ivan Oseledets. Lotr: Low tensor rank weight adaptation. 2024.
- [4] Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, Mansheej Paul, Philip Greengard, Connor Jennings, Daniel King, Sam Havens, Vitaliy Chiley, Jonathan Frankle, et al. Lora learns less and forgets less. *Transactions on Machine Learning Research*, 2024.
- [5] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems*, 36:10088–10115, 2023.
- [6] Ning Ding, Xingtai Lv, Qiaosen Wang, Yulin Chen, Bowen Zhou, Zhiyuan Liu, and Maosong Sun. Sparse low-rank adaptation of pre-trained language models. 2023.
- [7] Ning Ding, Yujia Qin, Guang Yang, Fuchao Wei, Zonghan Yang, Yusheng Su, Shengding Hu, Yulin Chen, Chi-Min Chan, Weize Chen, et al. Parameter-efficient fine-tuning of large-scale pre-trained language models. *Nature Machine Intelligence*, 5(3):220–235, 2023.
- [8] Abdessalam Ed-dib, Zhanibek Datbayev, and Amine Mohamed Abousalah. Gelora: Geometric adaptive ranks for efficient lora fine-tuning. *arXiv preprint arXiv:2412.09250*, 2024.
- [9] Vlad Fomenko, Han Yu, Jongho Lee, Stanley Hsieh, and Weizhu Chen. A note on lora. 2024.
- [10] Zihao Fu, Haoran Yang, Anthony Man-Cho So, Wai Lam, Lidong Bing, and Nigel Collier. On the effectiveness of parameter-efficient fine-tuning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pages 12799–12807, 2023.
- [11] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- [12] Yuxuan Hu, Jing Zhang, Xiaodong Chen, Zhe Zhao, Cuiping Li, and Hong Chen. Lors: Efficient low-rank adaptation for sparse large language model. *arXiv preprint arXiv:2501.08582*, 2025.
- [13] Hai Huang and Randall Balestriero. Allora: Adaptive learning rate mitigates lora fatal flaws. *arXiv preprint arXiv:2410.09692*, 2024.
- [14] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. *Advances in neural information processing systems*, 31, 2018.
- [15] Uijeong Jang, Jason D. Lee, and Ernest K. Ryu. Lora training in the ntk regime has no spurious local minima. 2024.
- [16] Yixin Ji, Yang Xiang, Juntao Li, Qingrong Xia, Zi Ye, Xinyu Duan, Zhefeng Wang, Kehai Chen, and Min Zhang. Adaptive feature-based low-rank compression of large language models via bayesian optimization. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 4152–4168, 2024.
- [17] Ting Jiang, Shaohan Huang, Shengyue Luo, Zihan Zhang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, Qi Zhang, Deqing Wang, and Fuzhen Zhuang. Mora: High-rank updating for parameter-efficient fine-tuning. 2024.
- [18] Omkar Khade, Shruti Jagdale, Abhishek Phaltankar, Gauri Takalikar, and Raviraj Joshi. Challenges in adapting multilingual llms to low-resource languages using lora peft tuning. *arXiv preprint arXiv:2411.18571*, 2024.
- [19] Qingyao Li, Lingyue Fu, Weiming Zhang, Xianyu Chen, Jingwei Yu, Wei Xia, Weinan Zhang, Ruiming Tang, and Yong Yu. Adapting large language models for education: Foundational capabilities, potentials, and challenges. *arXiv preprint arXiv:2401.08664*, 2023.
- [20] Yinghao Li, Vianne Gao, Chao Zhang, and MohamadAli Torkamani. Ensembles of low-rank expert adapters. 2025.
- [21] Xiaoxuan Liao, Chihang Wang, Shicheng Zhou, Jiacheng Hu, Hongye Zheng, and Jia Gao. Dynamic adaptation of lora fine-tuning for efficient and task-specific optimization of large language models. *arXiv preprint arXiv:2501.14859*, 2025.
- [22] Peiqin Lin, Shaoxiong Ji, Jörg Tiedemann, André FT Martins, and Hinrich Schütze. Mala-500: Massive language adaptation of large language models. *arXiv preprint arXiv:2401.13303*, 2024.
- [23] Jun Liu, Zhenglun Kong, Peiyan Dong, Xuan Shen, Pu Zhao, Hao Tang, Geng Yuan, Wei Niu, Wenbin Zhang, Xue Lin, et al. Rora: Efficient fine-tuning of llm with reliability optimization for rank adaptation. *arXiv preprint arXiv:2501.04315*, 2025.
- [24] Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. Spinqant: Llm quantization with learned rotations. *arXiv preprint arXiv:2405.16406*, 2024.
- [25] Zequan Liu, Jiawen Lyn, Wei Zhu, Xing Tian, and Yvette Graham. Alora: Allocating low-rank adaptation for fine-tuning large language models. 2024.
- [26] Jun Lu. Low-rank approximation, adaptation, and other tales. *arXiv preprint arXiv:2408.05883*, 2024.
- [27] Kai Lv, Yuqing Yang, Tengxiao Liu, Qinghui Gao, Qipeng Guo, and Xipeng Qiu. Full parameter fine-tuning for large language models with limited resources. *arXiv preprint arXiv:2306.09782*, 2023.
- [28] Deepak Narayanan, Mohammad Shoeybi, Jared Casper, Patrick LeGresley, Mostofa Patwary, Vijay Korthikanti, Dmitri Vainbrand, Prithvi Kashinkunti, Julie Bernauer, Bryan Catanzaro, et al. Efficient large-

- scale language model training on gpu clusters using megatron-lm. In *Proceedings of the international conference for high performance computing, networking, storage and analysis*, pages 1–15, 2021.
- [29] Humza Naveed, Asad Ullah Khan, Shi Qiu, Muhammad Saqib, Saeed Anwar, Muhammad Usman, Naveed Akhtar, Nick Barnes, and Ajmal Mian. A comprehensive overview of large language models. *arXiv preprint arXiv:2307.06435*, 2023.
- [30] Mahdi Nikdan, Soroush Tabesh, Elvir Crnčević, and Dan Alistarh. Rosa: Accurate parameter-efficient fine-tuning via robust adaptation. 2024.
- [31] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35:27730–27744, 2022.
- [32] Yehonathan Refael, Jonathan Svirskey, Boris Shustin, Wasim Huleihel, and Ofir Lindenbaum. Adarankgrad: Adaptive gradient-rank and moments for memory-efficient llms training and fine-tuning. *arXiv preprint arXiv:2410.17881*, 2024.
- [33] Pengjie Ren, Chengshun Shi, Shiguang Wu, Mengqi Zhang, Zhaochun Ren, Maarten de Rijke, Zhumin Chen, and Jiahuan Pei. Melora: Mini-ensemble low-rank adapters for parameter-efficient fine-tuning. 2024.
- [34] Rajarshi Saha, Naomi Sagan, Varun Srivastava, Andrea Goldsmith, and Mert Pilanci. Compressing large language models using low rank and low precision decomposition. *Advances in Neural Information Processing Systems*, 37:88981–89018, 2024.
- [35] Nikunj Saunshi, Sadhika Malladi, and Sanjeev Arora. A mathematical exploration of why language models help solve downstream tasks. *arXiv preprint arXiv:2010.03648*, 2020.
- [36] Ayan Sengupta, Vaibhav Seth, Arinjay Pathak, Natraj Raman, Sriram Gopalakrishnan, and Tanmoy Chakraborty. Robust and efficient fine-tuning of llms with bayesian reparameterization of low-rank adaptation. *arXiv preprint arXiv:2411.04358*, 2024.
- [37] Reece Shuttlesworth, Jacob Andreas, Antonio Torralba, and Pratyusha Sharma. Lora vs full fine-tuning: An illusion of equivalence. *arXiv preprint arXiv:2410.21228*, 2024.
- [38] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [39] Christophe Tribes, Sacha Benarroch-Lelong, Peng Lu, and Ivan Kobzyev. Hyperparameter optimization for large language model instruction-tuning. 2023.
- [40] Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu, Xifeng Yan, Jianfeng Gao, and Furu Wei. Augmenting language models with long-term memory. *Advances in Neural Information Processing Systems*, 36:74530–74543, 2023.
- [41] Chenxi Whitehouse, Fantine Huot, Jasmijn Bastings, Mostafa Dehghani, Chu-Cheng Lin, and Mirella Lapata. Low-rank adaptation for multilingual summarization: An empirical study. 2023.
- [42] Chao Xiao, Yifei Deng, Zhijie Yang, Renzhi Chen, Hong Wang, Jingyue Zhao, Huadong Dai, Lei Wang, Yuhua Tang, and Weixia Xu. Llm-based processor verification: A case study for neuronomorphic processor. In *2024 Design, Automation & Test in Europe Conference & Exhibition (DATE)*, pages 1–6. IEEE, 2024.
- [43] Xiaojun Xiao, Sen Shen, Qiming Bao, Hongfei Rong, Kairui Liu, Zhongsheng Wang, and Jiamou Liu. Cora: Optimizing low-rank adaptation with common subspace of large language models. *arXiv preprint arXiv:2409.02119*, 2024.
- [44] Adam X Yang, Maxime Robeyns, Xi Wang, and Laurence Aitchison. Bayesian low-rank adaptation for large language models. *arXiv preprint arXiv:2308.13111*, 2023.
- [45] Menglin Yang, Jialin Chen, Yifei Zhang, Jiahong Liu, Jiasheng Zhang, Qiyao Ma, Harshit Verma, Qianru Zhang, Min Zhou, Irwin King, et al. Low-rank adaptation for foundation models: A comprehensive review. *arXiv preprint arXiv:2501.00365*, 2024.
- [46] Yuchen Zeng and Kangwook Lee. The expressive power of low-rank adaptation. 2023.
- [47] Longteng Zhang, Lin Zhang, Shaohuai Shi, Xiaowen Chu, and Bo Li. Lora-fa: Memory-efficient low-rank adaptation for large language models fine-tuning. 2023.
- [48] Wenxuan Zhang, Yang Deng, and Wai Lam. Answer ranking for product-related questions via multiple semantic relations modeling. In *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval*, pages 569–578, 2020.
- [49] Changhai Zhou, Shijie Han, Shiyang Zhang, Shichao Weng, Zekai Liu, and Cheng Jin. Rankadaptor: Hierarchical dynamic low-rank adaptation for structural pruned llms. *arXiv preprint arXiv:2406.15734*, 2024.
- [50] Bojia Zi, Xianbiao Qi, Lingzhi Wang, Jianan Wang, Kam-Fai Wong, and Lei Zhang. Delta-lora: Fine-tuning high-rank parameters with the delta of low-rank matrices. 2023.